Intake Air Distribution and Filtering

Ford Ecat Electronic Parts Catalog

The intake air system consists of the:

- Engine Air Cleaner (ACL) .
- ACL element.
- ACL outlet pipe.
- Mass Air Flow (MAF) sensor.
- Manifold Absolute Pressure (MAP) sensor.
- Charge Air Cooler (CAC) - 3.5L Gasoline Turbocharged Direct Injection (GTDI) .
- CAC pipes and tubes - 3.5L GTDI .
- turbocharger intake pipes - 3.5L GTDI .
- turbocharger intake tubes - 3.5L GTDI .

The intake air system:

- cleans intake air with a replaceable, dry-type ACL element.
- measures airflow and air temperature with a MAF sensor. Refer to Section 303-14.
- measures manifold vacuum with a MAP sensor (3.5L GTDI engine) and converts it to an electronic signal. This provides the PCM information on the engine load. Refer to Section 303-14.

The engine ACL contains an ACL element made of treated, pleated paper. A new ACL element must be installed periodically as scheduled. Engine performance and fuel economy are adversely affected when maximum restriction of the ACL element is reached. The ACL housing is an integral part of the degas bottle (3.5L Gasoline Turbocharged Direct Injection (GTDI) , 3.7L and 5.0L engines) or the coolant expansion tank (6.2L engines) and cannot be serviced separately. Refer to Section 303-03.

The CAC subsystem (3.5L GTDI engine) cools and increases the density of the compressed turbocharged air.